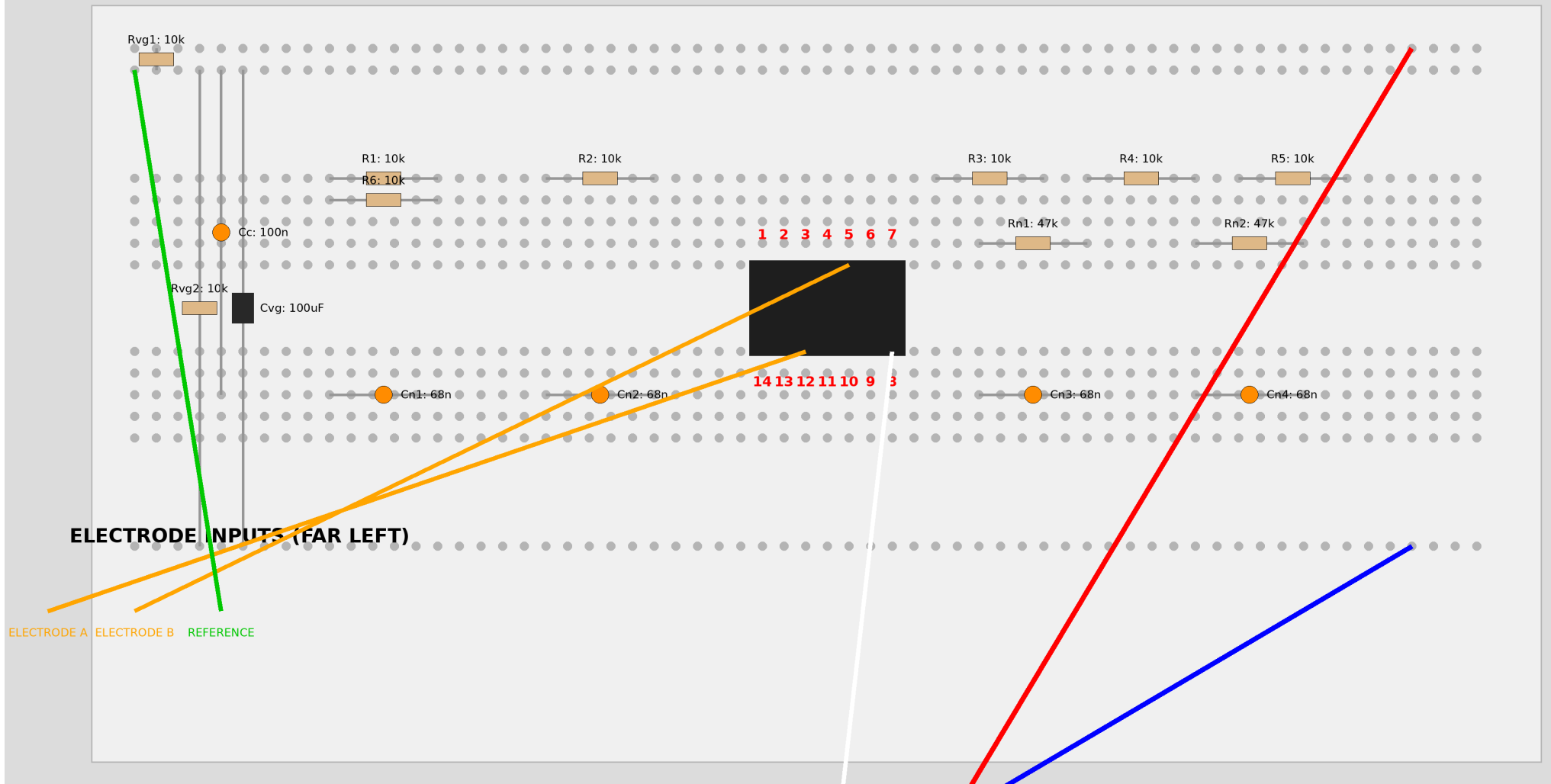


DIY EEG Circuit Guide (TL074 + Single 9V Battery)

EEG Circuit Schematic (Single TL074)

DIY EEG Circuit Guide (TL074 + Single 9V Battery)

ULTIMATE EEG LAYOUT - ALL COMPONENTS + SPATIAL CONSTRAINTS



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DESIGN.md

EEG Circuit Design (Single TL074, Single 9V Battery)

Architecture

1. Virtual Ground (VGND): 4.5V reference using two 10k resistors and a 100uF capacitor.
2. Stage 1: Instrumentation Amplifier: 3-op-amp InAmp, Gain ~ 21 .
3. Stage 2: 50Hz Notch Filter: Passive Twin-T, tuned for 49.8Hz.
4. Stage 3: Gain + Bandpass Filter: Gain ~ 471 , Bandpass 3.4Hz - 34Hz.

Total Gain: $\sim 10,000$.

DIY EEG Circuit Guide (TL074 + Single 9V Battery)

ASSEMBLY.md

Breadboard Assembly Guide

Step 1: Power & VGND

- Red Rail: +9V. Blue Rail: 0V.
- VGND: Junction of two 10k resistors between +9V and 0V. Stabilize with 100uF to 0V.

Step 2: TL074 IC

- Side 1 (Bottom): Pins 1-7. Side 2 (Top): Pins 14-8.
- Pin 4 to +9V. Pin 11 to 0V.

Step 3: Stages

- Stage 1 (InAmp): Pins 12, 13, 14, 5, 6, 7, 2, 3, 1.
- Stage 2 (Notch): Pin 1 to Input, Output to Pin 10.
- Stage 3: Output at Pin 8.

Step 4: Connections

- Electrodes: Left side of board.
- Battery: Under the board.
- Oscilloscope: Bottom of board.

TESTING.md

Safety & Testing

Safety

1. Battery Only.
2. Oscilloscope Isolation.
3. Reference Grounding.

Testing

- Check VGND (4.5V).
- Short inputs to VGND, check output (Flat line).
- Capture Alpha waves (8-13Hz) with eyes closed.

DIY EEG Circuit Guide (TL074 + Single 9V Battery)

CONNECTIONS.md

Detailed Connection List (Point-to-Point)

Power Supply & Virtual Ground

- Battery (+9V) -> Breadboard Red Rail (+)
- Battery (0V) -> Breadboard Blue Rail (-)
- Resistor Rvg1 (10k) -> Terminal 1: Red Rail (+), Terminal 2: VGND Rail
- Resistor Rvg2 (10k) -> Terminal 1: VGND Rail, Terminal 2: Blue Rail (-)
- Capacitor Cvg (100uF) -> Terminal (+): VGND Rail, Terminal (-): Blue Rail (-)

TL074 IC Connections

- Pin 4 (VCC+) -> Red Rail (+)
- Pin 11 (VCC-) -> Blue Rail (-)
- Pin 10 (Stage 3 Input+) -> VGND Rail (via 1M Rpd resistor)

Stage 1: Instrumentation Amplifier

- Rg (1k) -> Terminal 1: Pin 13, Terminal 2: Pin 6
- R1 (10k) -> Terminal 1: Pin 14, Terminal 2: Pin 13
- R2 (10k) -> Terminal 1: Pin 7, Terminal 2: Pin 6
- R3 (10k) -> Terminal 1: Pin 14, Terminal 2: Pin 2
- R4 (10k) -> Terminal 1: Pin 7, Terminal 2: Pin 3
- R5 (10k) -> Terminal 1: Pin 3, Terminal 2: VGND Rail
- R6 (10k) -> Terminal 1: Pin 2, Terminal 2: Pin 1

Stage 2: 50Hz Notch Filter

- Rn1 (47k) -> Terminal 1: Pin 1, Terminal 2: Node X

DIY EEG Circuit Guide (TL074 + Single 9V Battery)

- Rn2 (47k) -> Terminal 1: Node X, Terminal 2: Node Y (Notch Out)
- Cn1/Cn2 (2x 68n in parallel) -> From Node X to VGND Rail
- Cn3 (68n) -> Terminal 1: Pin 1, Terminal 2: Node Z
- Cn4 (68n) -> Terminal 1: Node Z, Terminal 2: Node Y (Notch Out)
- Rn3/Rn4 (2x 47k in parallel) -> From Node Z to VGND Rail

Stage 3: Gain & Filter

- Cc (100n) -> Terminal 1: Node Y (Notch Out), Terminal 2: Pin 10
- Rpd (1M) -> Terminal 1: Pin 10, Terminal 2: VGND Rail
- Rf (470k) -> Terminal 1: Pin 8, Terminal 2: Pin 9
- Cf (10n) -> Terminal 1: Pin 8, Terminal 2: Pin 9
- Rs (1k) -> Terminal 1: Pin 9, Terminal 2: Node W
- Cs (47uF) -> Terminal (+): Node W, Terminal (-): VGND Rail

External Interfaces

- Electrode A -> Pin 12
- Electrode B -> Pin 5
- Reference Electrode -> VGND Rail
- Oscilloscope Probe -> Pin 8
- Oscilloscope Ground -> Blue Rail (-) (Battery Negative)